

clean_data.ipynb ----- **Data Cleaning and Preprocessing**
knn.ipynb ----- **KNN and WNN Implementation**
prepare_dataset.py ----- **Dataset Preparation**
train_coral_cnn.py ----- **CNN Model Training**
XGBoost_hyperpara_featurescale.py ----- **XGBoost Hyperparameter Tuning**
[FinalXGBOOST.py](#) ----- **Final XGBoost Model**
predict_single_image.py ----- **Single Image Prediction**
evaluation [system.py](#) ----- **Model Evaluation**